
Appendix A - Training

Includes:

Operator Training Guide

Maintenance Training Guide



Operator Training

Trainee prerequisites: Knowledge of basic mechanical operations and reading.
 Classroom hours: ½ hour
 Hand-on-training: 3½ hours
 Total training hours: 4 hours

Trainee Materials: Safety Guide (pages 13-15 of this manual) and Operator's Guide (pages 21-34 of this manual)

Task or Teaching Point	Additional Information / Suggestions
<u>CLASSROOM</u>	
I. INTRODUCTION	
A. Briefly describe the course and what each trainee will learn.	
II. MANUAL TRAINING	Service Manual - You may want to run copies or print the appropriate pages to hand out.
A. Controls	Pg. 26
B. Pre-Operation	Pg. 24
C. Start-Up	Pg. 24
D. Normal Shutdown	Pg. 24
E. Emergency Shutdown	Pg. 25
F. Changeover	Pg. 25 and Pg. 32
G. Change Existing Recipe / Parameters	Pg. 25, and Pgs. 27-29
H. Single Lane Clearout	Pg. 25
I. Clearing Jams	Pg. 25
<u>HANDS-ON-TRAINING</u>	
I. NORMAL MACHINE OPERATION	
A. Demonstrate Startup	
B. Run product and discuss the machine processes.	
1. Count PE	
2. Stacking	
C. Demonstrate Shutdown	
D. Demonstrate Changeover	
1. Change Recipe	
2. Change Insert Tray	
3. Infeed Alignment Gates	

Operator Training

Task or Teaching Point	Additional Information / Suggestions
II. CHANGEOVER A. Demonstrate the changeover procedure carefully explaining each adjustment. 1. Change recipe. 2. Change Insert Tray 3. Infeed Alignment Gates B. Startup and run 1 or 2 products to verify correct adjustments.	Be sure to have 2 different tortilla sizes for demonstration purposes. You should be sure to explain all adjustments, whether they are to be performed for this changeover or not. Allow a trainee to startup the machine.
III. JAMS A. Start the machine. B. Run product and demonstrate the proper steps to clear the jam. C. Repeat 'B' and allow the operator(s) to clear the jam. D. Discuss what caused the jam and the proper steps to prevent future jamming. E. Shut the machine down (if it is running).	Set the machine up to jam before proceeding. You may want to let a trainee start the machine at this point. Allow a trainee to shut the machine down.
IV. DAILY MAINTENANCE A. Wipe Down the Machine 1. Dust 2. Product Accumulation 3. Grease B. Check Safety Switches 1. Stop Button 2. All Guard Doors (optional)	

Operator Performance Assessment

Before the Performance Assessment

1. Verify that the work area conforms to all safety standards.
2. A different product size from the one the machine is currently set up for, should be available to the trainee.

During the Performance Assessment

1. Stress the importance of following all safety standards.
2. Ask the trainee to complete each task on the checklist.
3. Check each block of the performance checklist as the trainee satisfactorily performs the step.
4. If the trainee fails to complete a task satisfactorily:
 - a. End the evaluation.
 - b. Review the correct procedure with the trainee.
 - c. Schedule a new assessment time for the trainee.

After the Performance Assessment

1. Verify that the work area has been cleaned up.
2. Verify that the work area is safe.
3. Review the Performance Checklist with the trainee.
4. Sign and date the Performance Checklist on the line provided.
5. Have the trainee sign and date the Performance Checklist on the line provided.
7. Forward all records to the appropriate department, for record maintenance (ie. Training Department, Personnel...)



Operator Performance Assessment

<u>Tasks</u>	<u>Sat</u>	<u>Unsat</u>
1. Change product recipe.		
2. Install insert tray.		
3. Adjust infeed alignment guides.		
Clear a jam.		
1. Push the Emergency Stop button or open guard.		
2. Remove any product jammed in the counter stacker.		
3. Inspect for equipment damage or debris fallout.		
4. Account for all tools, parts, or materials.		
5. Close guard.		
6. Reset the Emergency Stop button, if necessary.		
7. Push the Safety Reset button.		
8. Push the Cycle Start button.		
9. Check the next five stacks to make sure the machine is operating correctly.		
Changeover.		
1. Push the Cycle Stop button (to allow product out of the stacking area).		
2. Push the Emergency Stop button.		
3. Replace Insert Tray (if required).		
4. Adjust Infeed Alignment Guides (if required).		
5. Change product recipe (on display).		
6. Account for all tools, parts, or materials		
7. Reset the EMERGENCY STOP button.		
8. Push the Safety Reset button.		
9. Push the Cycle Start button.		
10. Check the next five stacks to make sure the machine is operating correctly.		
Comments		

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Operator Performance Assessment

Evaluator Signature and Date
Trainee Signature and Date
Trainee Signature and Date
Trainee Signature and Date
Trainee Signature and Date
Trainee Signature and Date
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Trainee Signature and Date
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Maintenance Training

Trainee prerequisites: Completion of Operator Training, Knowledge of basic maintenance operations, and reading.
 Classroom hours: ¾ hour
 Hands-on-training: 3 ¼ hours
 Total training hours: 4 hours

Instructor Materials: Two (2) Locks with keys for Air Dump and Main Electrical Panel, Screw Driver Set, Volt Meter, Basic Tool Kit, 1/4-3/8" driver set.

Trainee Materials: Machine Manual for Arr-Tech Counter Stacker Serial# A17006 and Vendor documentation showing maintenance information for machine components.

Task or Teaching Point	Additional Information / Suggestions
<u>CLASSROOM</u>	
I. INTRODUCTION	
A. Briefly describe the course and what each trainee will learn.	
II. MANUAL TRAINING	Service Manual - You may want to run copies or print the appropriate pages to hand out.
A. User Interface	Pgs. 27-31
B. Lubrication Fluids	Pg. 30-31
C. Daily Maintenance	Pg. 60
D. Weekly Maintenance	Pg. 61
E. Monthly Maintenance	Pg. 62
F. Semi Annual Maintenance	Pg. 63
<u>HANDS-ON-TRAINING</u>	
I. LOCK OUT PROCEDURES	Each task should be demonstrated for the trainees and whenever possible the trainees should be allowed to perform the tasks.
A. Panel	
B. Air System	
II. DAILY MAINTENANCE	
A. Clean the machine.	
B. Inspect/replace mechanical components.	
C. Inspect/adjust belt tension (infeed, discharge, conveyor).	
D. Inspect/test sensors.	
E. Inspect/drain air filter.	
F. Clean servo system.	
G. Check Safety Switches.	
1. Emergency Stop Buttons	
2. Guard Doors	

Maintenance Training

Task or Teaching Point	Additional Information / Suggestions
III. WEEKLY MAINTENANCE A. Check vibration area. B. Check air driven device operation (forks). C. Grease bearings. D. Check/replace red urethane belts. E. Tighten bearing set screws. F. Check/replace motors and bearings. G. Inspect servo motors and cables. Repair or replace if needed.	
IV. MONTHLY MAINTENANCE A. Check/adjust conveyor belt tension. B. Check/tighten fasteners. C. Inspect/repair pneumatic system. D. Check/fill main drive gear box oil. E. Check amp draw on all motors. F. Check full load amp draw at main power. G. Inspect all electrical connections. H. Inspect air chute blower motor and clean intake. I. Remove air chute back and vacuum all dust from air chute.	Review locations of all electrical connections.
V. SEMI-ANNUAL MAINTENANCE A. Inspect/repair/replace grip top infeed conveyor belt. B. Inspect rod and cylinder jam nuts (interrupt forks) for tightness. C. Check/replace cylinder nose bushing for wear/blow-by. D. Inspect/replace electrical cords, conduit, conduit fittings, conduit boxes and conduit compression fittings.	
VI. AIR FILTER REGULATOR MAINTENANCE A. O-Ring Replacement B. Air Filter C. Soft Start Module	

Maintenance Performance Assessment

Before the Performance Assessment

1. Verify that the work area conforms to all safety standards.

During the Performance Assessment

1. Stress the importance of following all safety standards.
2. Ask the trainee to complete each task on the checklist.
Note: The machine may not require the completion of some maintenance tasks. In the case of these tasks the trainee should be able to demonstrate the ability to complete the task, by explaining the procedure or going through the motions as if completing the task.
3. Check each block of the performance checklist as the trainee satisfactorily performs or satisfactorily demonstrates the ability to perform the step.
4. If the trainee fails to complete a task satisfactorily:
 - a. End the evaluation.
 - b. Review the correct procedure with the trainee.
 - c. Schedule a new assessment time for the trainee.

After the Performance Assessment

1. Verify that the work area has been cleaned up.
2. Verify that the work area is safe.
3. Review the Performance Checklist with the trainee.
4. Sign and date the Performance Checklist on the line provided.
5. Have the trainee sign and date the Performance Checklist on the line provided.
7. Forward all records to the appropriate department, for record maintenance (ie. Training Department, Personnel...)

Maintenance Performance Assessment

<u>Tasks</u>	<u>Sat</u>	<u>Unsat</u>
Lock Out Procedures		
1. Electrical System (Panel)		
2. Air System		
Daily Maintenance		
1. Clean the machine.		
2. Inspect/replace mechanical components		
3. Inspect/adjust belt tension (infeed, discharge, conveyor).		
4. Inspect/test sensors		
5. Inspect/drain air filter		
6. Clean servo system.		
7. Check operation of Emergency Stop buttons.		
8. Check operation of Guard Door sensors.		
8. Check operation of Guard Door sensors.		
Weekly Maintenance		
1. Check vibration area.		
2. Check air driven device operation (forks).		
3. Grease bearings.		
4. Check/replace red urethane belts.		
5. Tighten bearing set screws.		
6. Check/replace motors/bearings.		
7. Inspect servo motors and cables. Repair or replace if needed.		
Monthly Preventative Maintenance		
1. Check/adjust conveyor belt tension.		
2. Check/tighten fasteners.		
3. Inspect/repair pneumatic system.		
4. Check/fill main drive gear box oil.		
5. Check amp draw on all motors.		
6. Check full load amp draw at main power.		
7. Inspect all electrical connections.		
8. Inspect air chute blower motor and clean intake.		
9. Remove air chute back and vacuum all dust from air chute.		
Semi-annual Maintenance		
1. Inspect/repair/replace grip top infeed conveyor belt.		
2. Inspect rod and cylinder jam nuts (interrupt forks) for tightness.		
3. Check/replace cylinder nose bushing for wear/blow-by.		
4. Inspect/replace electrical cords, conduit, conduit fittings, conduit boxes and conduit compression fittings.		

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Maintenance Performance Assessment

<u>Tasks</u>	<u>Sat</u>	<u>Unsat</u>
Air/Filter/Regulator Maintenance		
1. Replace O-Ring		
2. Clean air filter.		
1. Clean soft start module.		
Comments		
Evaluator Signature and Date		
Trainee Signature and Date		
Trainee Signature and Date		
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